Worksheet: The Cost of Making a Car

ECON 201

Definitions

Cost function, $C(Q)$: the total cost of producing Q units of output. Suppose a firm has fixed costs F, and its variable costs are directly proportional to the quantity it produces. Then the total cost of producing Q units is

$$C(Q) = F + cQ$$

where F is the **fixed cost**, paid regardless of what Q is, and cQ is the **variable cost**: c per unit, times Q units.

Average cost, AC: the cost per unit,

$$\text{AC} = \frac{C(Q)}{Q}$$

Marginal cost, MC: the increase in total cost from producing one more unit,

$$\text{MC} = \frac{\Delta C}{\Delta Q}$$

where Δ means "the change in": ΔC is the change in total cost when output changes by ΔQ.

1. Fixed or variable?

Beautiful Cars is a small firm that manufactures specialty cars. It has to pay for the following inputs to make its cars. Select all the costs that change with the number of cars the firm makes each day.

- ☐ A factory equipped with machinery to assemble the cars.
- ☐ Raw materials and components.
- ☐ Production workers to operate the equipment.
- ☐ Other workers to manage the production process, and to market and sell the finished cars.
- ☐ Research and development to develop future models.

2. The cost function of Beautiful Cars

Beautiful Cars has fixed costs of \$60,000 per day, and each extra car costs \$10,000, so its cost function is

$$C(Q) = 60{,}000 + 10{,}000\,Q$$

Fill in total cost, average cost, and marginal cost at each output. For marginal cost, use the cost of one more car.

Cars per day, Q	10	20	50
Total cost, $C(Q)$			
Average cost, AC			
Marginal cost, MC			

Average cost falls as Q rises, while marginal cost does not change. Explain why in a sentence or two.